

BEE 4750 Homework 4: Linear Programming and Capacity Expansion

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Due Date

Thursday, 11/06/25, 9:00pm

Overview

Instructions

- Problem 1 asks you to analytically (or graphically) solve a linear program.
- Problem 2 asks you to formulate and solve a resource allocation problem using linear programming.
- Problem 3 asks you to formulate, solve, and analyze a standard generating capacity expansion problem.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

Activating project at `~/BEE\_4750/hw4-evelyn-natalie`

```

using JuMP
using HiGHS
using DataFrames
using Plots
using Measures
using CSV
using MarkdownTables

```

Problems (Total: 30 Points)

Problem 1 (Total: 3 Points)

Solve the following linear program **analytically**. Show your work and justify any reasoning.

$$\begin{aligned}
 & \max_{x,y} && 3x + 2y \\
 & \text{subject to:} && x + 4y \leq 16 \\
 & && x - 2y \leq -1 \\
 & && 0 \leq x \leq 4 \\
 & && 1 \leq y \leq 4.
 \end{aligned}$$

Check attached!

Problem 2 (12 points)

A farmer has access to a pesticide which can be used on corn, soybeans, and wheat fields, and costs \$70/kg-yr to apply. The crop yields the farmer can obtain following crop yields by applying varying rates of pesticides to the field are shown in Table 1.

Application Rate (kg/ha)	Soybean (kg/ha)	Wheat (kg/ha)	Corn (kg/ha)
0	2900	3500	5900
1	3800	4100	6700
2	4400	4200	7900

Table 1: Crop yields from applying varying pesticide rates for Problem 1.

The costs of production, *excluding pesticides*, for each crop, and selling prices, are shown in Table 2.

Crop	Production Cost (\$/ha-yr)	Selling Price (\$/kg)
Soybeans	350	0.36
Wheat	280	0.27
Corn	390	0.22

Table 2: Costs of crop production, excluding pesticides, and selling prices for each crop.

Recently, environmental authorities have declared that farms cannot have an *average* application rate on soybeans, wheat, and corn which exceeds 0.8, 0.7, and 0.6 kg/ha, respectively. The farmer has asked you for advice on how they should plant crops and apply pesticides to maximize profits over 130 total ha while remaining in regulatory compliance if demand for each crop (which is the maximum the market would buy) this year is 250,000 kg?

Problem 2.1

Formulate a linear program for this resource allocation problem, including clear definitions of decision variable(s) (including units), objective function(s), and constraint(s) (make sure to explain functions and constraints with any needed derivations and explanations). **Tip: Make sure that all of your constraints are linear.**

Problem 2.2

How many ha should the farmer dedicate to each crop and with what pesticide application rate(s)? How much profit will the farmer expect to make?

Problem 2.3

The farmer has an opportunity to buy an extra 10 ha of land. How much extra profit would this land be worth to the farmer? Discuss why this value makes sense and under what conditions you would recommend the farmer should make the purchase.

Problem 2.1 Solutions

```
farmer_model = Model(HiGHS.Optimizer) # initialize model object

# Decision variables
# A[c,r] = hectares of land allocated to crop c using pesticide rate r (kg/ha·yr)
# Crops: Soybeans, Wheat, Corn
# Rates: 0, 1, 2 kg/ha·yr
```

```

@variable(farmer_model, A[1:9] >= 0) # vector index:
# 1: Soy@0, 2: Soy@1, 3: Soy@2,
# 4: Wheat@0, 5: Wheat@1, 6: Wheat@2,
# 7: Corn@0, 8: Corn@1, 9: Corn@2

# Profit = revenue - production cost - pesticide cost
# Soybean profits:
# rate 0:  $2900 \times 0.36 - 350 - 0 \times 70 = 694$ 
# rate 1:  $3800 \times 0.36 - 350 - 1 \times 70 = 948$ 
# rate 2:  $4400 \times 0.36 - 350 - 2 \times 70 = 1094$ 
#
# Wheat profits:
# rate 0:  $3500 \times 0.27 - 280 - 0 \times 70 = 665$ 
# rate 1:  $4100 \times 0.27 - 280 - 1 \times 70 = 757$ 
# rate 2:  $4200 \times 0.27 - 280 - 2 \times 70 = 714$ 
#
# Corn profits:
# rate 0:  $5900 \times 0.22 - 390 - 0 \times 70 = 908$ 
# rate 1:  $6700 \times 0.22 - 390 - 1 \times 70 = 1014$ 
# rate 2:  $7900 \times 0.22 - 390 - 2 \times 70 = 1208$ 

@objective(farmer_model, Max,
  694*A[1] + 948*A[2] + 1094*A[3] + # Soy
  665*A[4] + 757*A[5] + 714*A[6] + # Wheat
  908*A[7] + 1014*A[8] + 1208*A[9]) # Corn

# Total land constraint (ha)
@constraint(farmer_model, land,
  A[1] + A[2] + A[3] +
  A[4] + A[5] + A[6] +
  A[7] + A[8] + A[9] <= 130)
# Demand constraints (kg maximum)
@constraint(farmer_model, demand_soy,
  2900*A[1] + 3800*A[2] + 4400*A[3] <= 250000)
@constraint(farmer_model, demand_wheat,
  3500*A[4] + 4100*A[5] + 4200*A[6] <= 250000)
@constraint(farmer_model, demand_corn,
  5900*A[7] + 6700*A[8] + 7900*A[9] <= 250000)

```

```

# Average pesticide use constraints (linear)
# Soy: (1*A2 + 2*A3) 0.8*(A1 + A2 + A3)
@constraint(farmer_model, avg_soy,
    1*A[2] + 2*A[3] <= 0.8*(A[1] + A[2] + A[3]))
# Wheat: (1*A5 + 2*A6) 0.7*(A4 + A5 + A6)
@constraint(farmer_model, avg_wheat,
    1*A[5] + 2*A[6] <= 0.7*(A[4] + A[5] + A[6]))
# Corn: (1*A8 + 2*A9) 0.6*(A7 + A8 + A9)
@constraint(farmer_model, avg_corn,
    1*A[8] + 2*A[9] <= 0.6*(A[7] + A[8] + A[9]))
print(farmer_model)

```

```

Max 694 A[1] + 948 A[2] + 1094 A[3] + 665 A[4] + 757 A[5] + 714 A[6] + 908 A[7] + 1014 A[8] + 1014 A[9]
Subject to
    land : A[1] + A[2] + A[3] + A[4] + A[5] + A[6] + A[7] + A[8] + A[9] 130
    demand_soy : 2900 A[1] + 3800 A[2] + 4400 A[3] 250000
    demand_wheat : 3500 A[4] + 4100 A[5] + 4200 A[6] 250000
    demand_corn : 5900 A[7] + 6700 A[8] + 7900 A[9] 250000
    avg_soy : -0.8 A[1] + 0.19999999999999996 A[2] + 1.2 A[3] 0
    avg_wheat : -0.7 A[4] + 0.30000000000000004 A[5] + 1.3 A[6] 0
    avg_corn : -0.6 A[7] + 0.4 A[8] + 1.4 A[9] 0
    A[1] 0
    A[2] 0
    A[3] 0
    A[4] 0
    A[5] 0
    A[6] 0
    A[7] 0
    A[8] 0
    A[9] 0

```

In this part, I set up a linear program to help the farmer decide how to allocate land among soybeans, wheat, and corn while keeping pesticide use within regulatory limits. The decision variables represent both the area of land (in hectares) assigned to each crop and the pesticide application rate (in kg/ha).

The objective function maximizes total profit, calculated as total revenue minus production and pesticide costs. The model includes constraints for: Total land availability (limited to 130 ha) Maximum market demand for each crop (250,000 kg) Average pesticide limits (0.8, 0.7, and 0.6 kg/ha for soybeans, wheat, and corn) Non-negativity of all decision variables

The printed model confirms all constraints are linear and properly defined. This formulation sets the foundation for the next parts, where the model is solved to find the optimal land distribution and profit.

Problem 2.2 Solutions

The optimal plan dedicates 69.06 ha to soybeans, 22.48 ha to wheat, and 38.46 ha to corn. Within those totals, soybeans use 13.81 ha at the low rate, 55.25 ha at the medium rate, and 0 ha at the high rate; wheat uses 15.73 ha at the low rate, 0 ha at the medium rate, and 6.74 ha at the high rate; corn uses 26.92 ha at the low rate, 0 ha at the medium rate, and 11.54 ha at the high rate. This allocation satisfies all constraints and yields an expected profit of \$116,741.17.

```
# Solve the model
optimize!(farmer_model)

# Print model
@show termination_status(farmer_model)

# Decision variable values: A[1:9]
# Index meaning:
# 1:Soy@0, 2:Soy@1, 3:Soy@2, 4:Wheat@0, 5:Wheat@1, 6:Wheat@2, 7:Wheat@2, 8:Wheat@2, 9:Wheat@2
@show value.(A);

# Optimal profit
@show objective_value(farmer_model);

# Shadow prices
@show shadow_price(land);
@show shadow_price(demand_soy);
@show shadow_price(demand_wheat);
@show shadow_price(demand_corn);
@show shadow_price(avg_soy);
@show shadow_price(avg_wheat);
@show shadow_price(avg_corn);

soy_total = sum(value.(A[1:3]))
```

```
wheat_total = sum(value.(A[4:6]))
corn_total  = sum(value.(A[7:9]))
println("\nTotals by crop (ha):")
println("Soybeans = ", round(soy_total, digits=6))
println("Wheat = ", round(wheat_total, digits=6))
println("Corn = ", round(corn_total, digits=6))
```

Running HiGHS 1.11.0 (git hash: 364c83a51e): Copyright (c) 2025 HiGHS under MIT licence terms

LP has 7 rows; 9 cols; 27 nonzeros

Coefficient ranges:

Matrix [2e-01, 8e+03]

Cost [7e+02, 1e+03]

Bound [0e+00, 0e+00]

RHS [1e+02, 2e+05]

Presolving model

7 rows, 9 cols, 27 nonzeros 0s

7 rows, 9 cols, 27 nonzeros 0s

Presolve : Reductions: rows 7(-0); columns 9(-0); elements 27(-0) - Not reduced

Problem not reduced by presolve: solving the LP

Using EKK dual simplex solver - serial

Iteration	Objective	Infeasibilities	num(sum)
0	-1.8170297740e+02	Ph1: 7(28.3601); Du: 9(181.703)	0s
7	1.1674116702e+05	Pr: 0(0)	0s

Model status : Optimal

Simplex iterations: 7

Objective value : 1.1674116702e+05

P-D objective error : 6.2325284535e-17

HiGHS run time : 0.01

termination_status(farmer_model) = MathOptInterface.OPTIMAL

value.(A) = [13.812154696132604, 55.24861878453038, 0.0, 6.743306417339563, 15.7343816404589,

objective_value(farmer_model) = 116741.16702082448

shadow_price(land) = 729.4

shadow_price(demand_soy) = 0.046353591160221

shadow_price(demand_wheat) = -0.0

shadow_price(demand_corn) = 0.04132307692307692

shadow_price(avg_soy) = 212.28176795580106

shadow_price(avg_wheat) = 92.00000000000004

shadow_price(avg_corn) = 108.67692307692309

Totals by crop (ha):
Soybeans = 69.060773
Wheat = 22.477688
Corn = 38.461538

Problem 2.3 Solutions

If the farmer has the opportunity to buy an extra 10 hectares of land, the model shows that this would increase profit by about \$7,294. This value comes from the shadow price of land, which was \$729.4 per hectare. When I re-solved the model with total land increased from 130 ha to 140 ha, the total profit rose from \$116,741.17 to \$124,035.17, matching the shadow price estimate almost exactly.

This makes sense because land is currently the limiting resource in the model all 130 ha are already being used, so every additional hectare directly adds to total production and profit. I would recommend that the farmer purchase the extra land if the cost of acquiring and operating it is less than \$7,294 total (or roughly \$729 per hectare). If land prices are higher than that, the investment would not be justified under current market and regulatory conditions. However, if demand or prices for the crops increase, the value of additional land would likely rise as well, making the purchase a cost effective investment.

```
# Marginal value estimate from shadow price
@show shadow_price(land)
extra_profit_estimate = 10.0 * shadow_price(land)
@show extra_profit_estimate

# 2) Re-solve with same equations using total land increased to 140 ha
m2 = Model(HiGHS.Optimizer)
@variable(m2, A2[1:9] >= 0)

# Objective
@objective(m2, Max,
    694*A2[1] + 948*A2[2] + 1094*A2[3] +      # Soy
    665*A2[4] + 757*A2[5] + 714*A2[6] +      # Wheat
    908*A2[7] + 1014*A2[8] + 1208*A2[9] )    # Corn

# Constraints (same as shown in 2.3, except now the land RHS = 140)
@constraint(m2, land2,
    A2[1]+A2[2]+A2[3] + A2[4]+A2[5]+A2[6] + A2[7]+A2[8]+A2[9] <= 140)
```

```

@constraint(m2, demand_soy2,
2900*A2[1] + 3800*A2[2] + 4400*A2[3] <= 250000)
@constraint(m2, demand_wheat2,
3500*A2[4] + 4100*A2[5] + 4200*A2[6] <= 250000)
@constraint(m2, demand_corn2,
5900*A2[7] + 6700*A2[8] + 7900*A2[9] <= 250000)
@constraint(m2, avg_soy2,
1*A2[2] + 2*A2[3] <= 0.8*(A2[1] + A2[2] + A2[3]))
@constraint(m2, avg_wheat2,
1*A2[5] + 2*A2[6] <= 0.7*(A2[4] + A2[5] + A2[6]))
@constraint(m2, avg_corn2,
1*A2[8] + 2*A2[9] <= 0.6*(A2[7] + A2[8] + A2[9]))
optimize!(m2)

# Comparing profits
@show objective_value(farmer_model)    # 130 ha
@show objective_value(m2)              # with an increase to 140 ha
extra_profit_exact = objective_value(m2) - objective_value(farmer_model)
@show extra_profit_exact

```

```

shadow_price(land) = 729.4
extra_profit_estimate = 7294.0
Running HiGHS 1.11.0 (git hash: 364c83a51e): Copyright (c) 2025 HiGHS under MIT licence terms
LP has 7 rows; 9 cols; 27 nonzeros
Coefficient ranges:
  Matrix [2e-01, 8e+03]
  Cost    [7e+02, 1e+03]
  Bound   [0e+00, 0e+00]
  RHS     [1e+02, 2e+05]
Presolving model
7 rows, 9 cols, 27 nonzeros 0s
7 rows, 9 cols, 27 nonzeros 0s
Presolve : Reductions: rows 7(-0); columns 9(-0); elements 27(-0) - Not reduced
Problem not reduced by presolve: solving the LP
Using EKK dual simplex solver - serial
  Iteration      Objective      Infeasibilities num(sum)
          0      -1.8170297740e+02 Ph1: 7(28.3601); Du: 9(181.703) 0s
          7       1.2403516702e+05 Pr: 0(0) 0s
Model status      : Optimal
Simplex iterations: 7
Objective value    : 1.2403516702e+05

```

```
P-D objective error : 0.0000000000e+00
HiGHS run time      : 0.00
objective_value(farmer_model) = 116741.16702082448
objective_value(m2) = 124035.16702082449
extra_profit_exact = 7294.000000000015
```

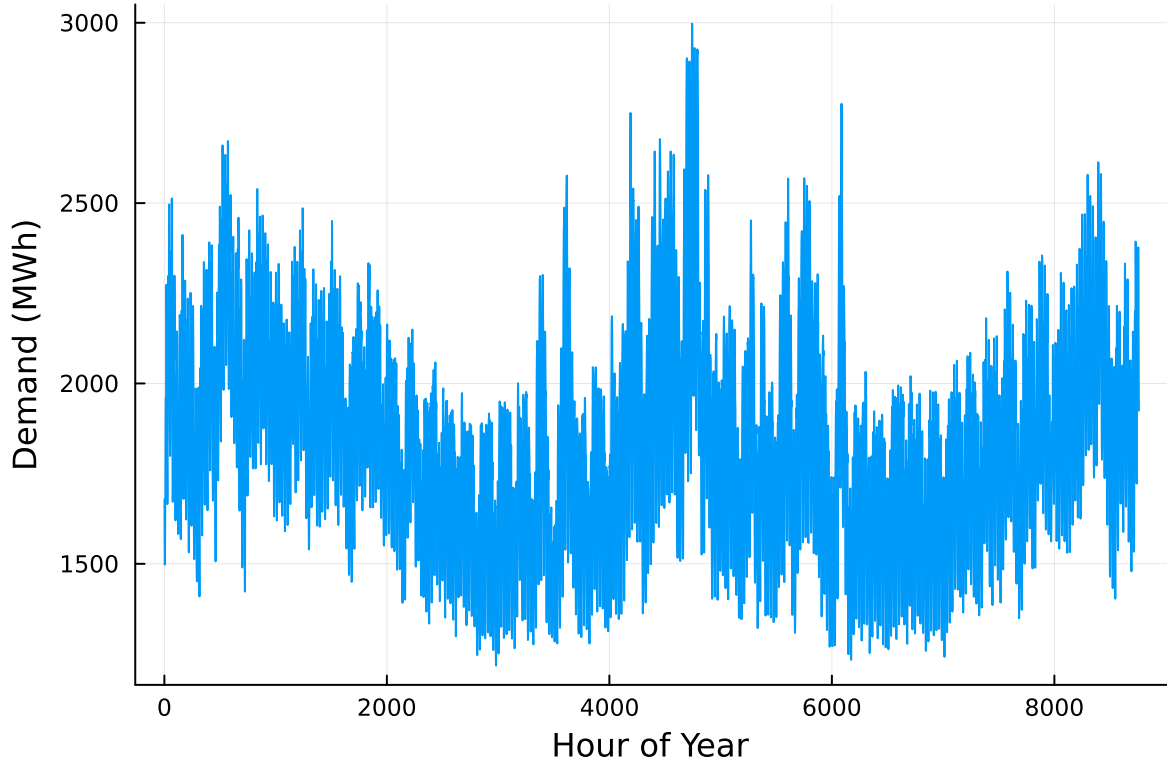
```
7294.000000000015
```

Problem 3 (15 points)

For this problem, we will use hourly load (demand) data from 2013 in New York's Zone C (which includes Ithaca). The load data is loaded and plotted below in Figure 1.

```
# load the data, pull Zone C, and reformat the DataFrame
NY_demand = DataFrame(CSV.File("data/2013_hourly_load_NY.csv"))
rename!(NY_demand, : "Time Stamp" => :Date)
demand = NY_demand[:, [:Date, :C]]
rename!(demand, :C => :Demand)
demand[:, :Hour] = 1:nrow(demand)

# plot demand
plot(demand.Hour, demand.Demand, xlabel="Hour of Year", ylabel="Demand (MWh)", label=:false)
```



Next, we load the generator data, shown in Table 3. This data includes fixed costs (\$/MW installed), variable costs (\$/MWh generated), and CO2 emissions intensity (tCO2/MWh generated).

```
gens = DataFrame(CSV.File("data/generators.csv"))
```

	Plant	FixedCost	VarCost	Emissions
	String15	Int64	Int64	Float64
1	Geothermal	450000	0	0.0
2	Coal	220000	24	1.0
3	NG CCGT	82000	30	0.43
4	NG CT	65000	40	0.55
5	Wind	91000	0	0.0
6	Solar	70000	0	0.0

Finally, we load the hourly solar and wind capacity factors, which are plotted in Figure 2. These tell us the fraction of installed capacity which is expected to be available in a given hour for generation (typically based on the average meteorology).

```

# load capacity factors into a DataFrame
cap_factor = DataFrame(CSV.File("data/wind_solar_capacity_factors.csv"))

# plot January capacity factors
p1 = plot(cap_factor.Wind[1:(24*31)], label="Wind")
plot!(cap_factor.Solar[1:(24*31)], label="Solar")
xaxis!("Hour of the Month")
yaxis!("Capacity Factor")

p2 = plot(cap_factor.Wind[4344:4344+(24*31)], label="Wind")
plot!(cap_factor.Solar[4344:4344+(24*31)], label="Solar")
xaxis!("Hour of the Month")
yaxis!("Capacity Factor")

display(p1)
display(p2)

```

You have been asked to develop a generating capacity expansion plan for the utility in Riley County, NY, which currently has no existing electrical generation infrastructure. The utility can build any of the following plant types: geothermal, coal, natural gas combined cycle gas turbine (CCGT), natural gas combustion turbine (CT), solar, and wind. The NY state legislature is planning to enact an annual CO₂ limit, which for the utility would limit the emissions in its footprint to 1.5 MtCO₂/yr.

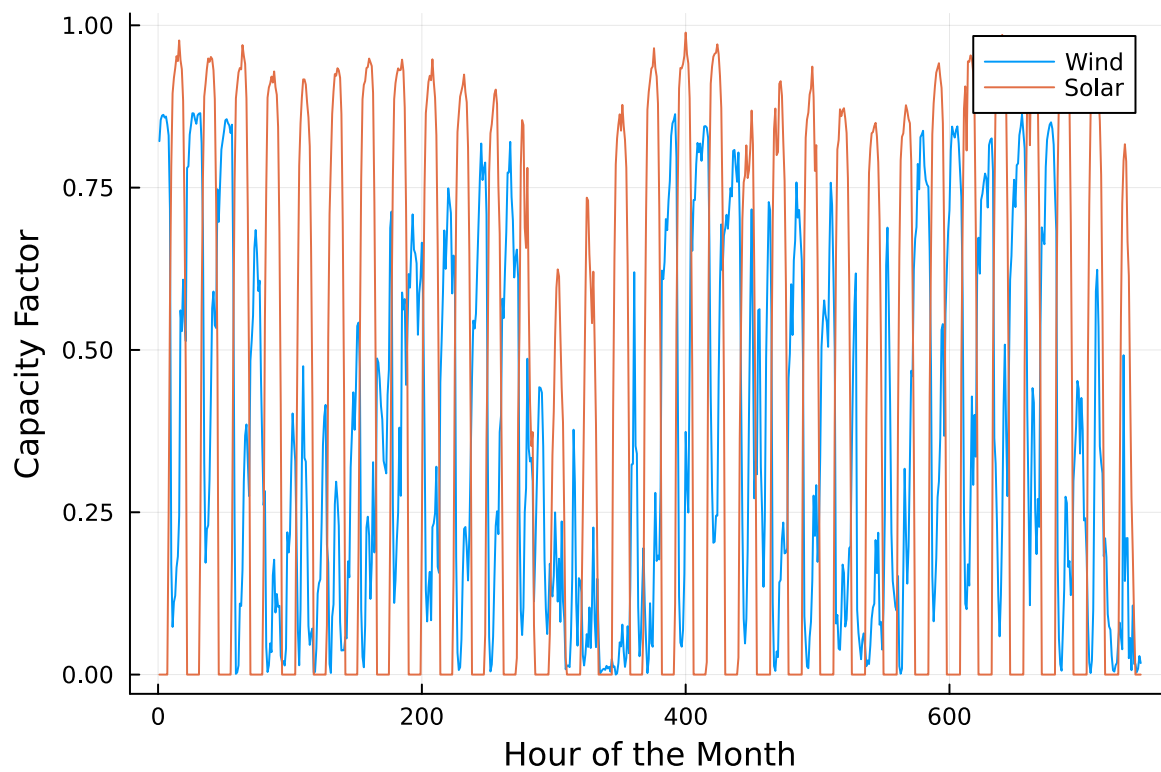
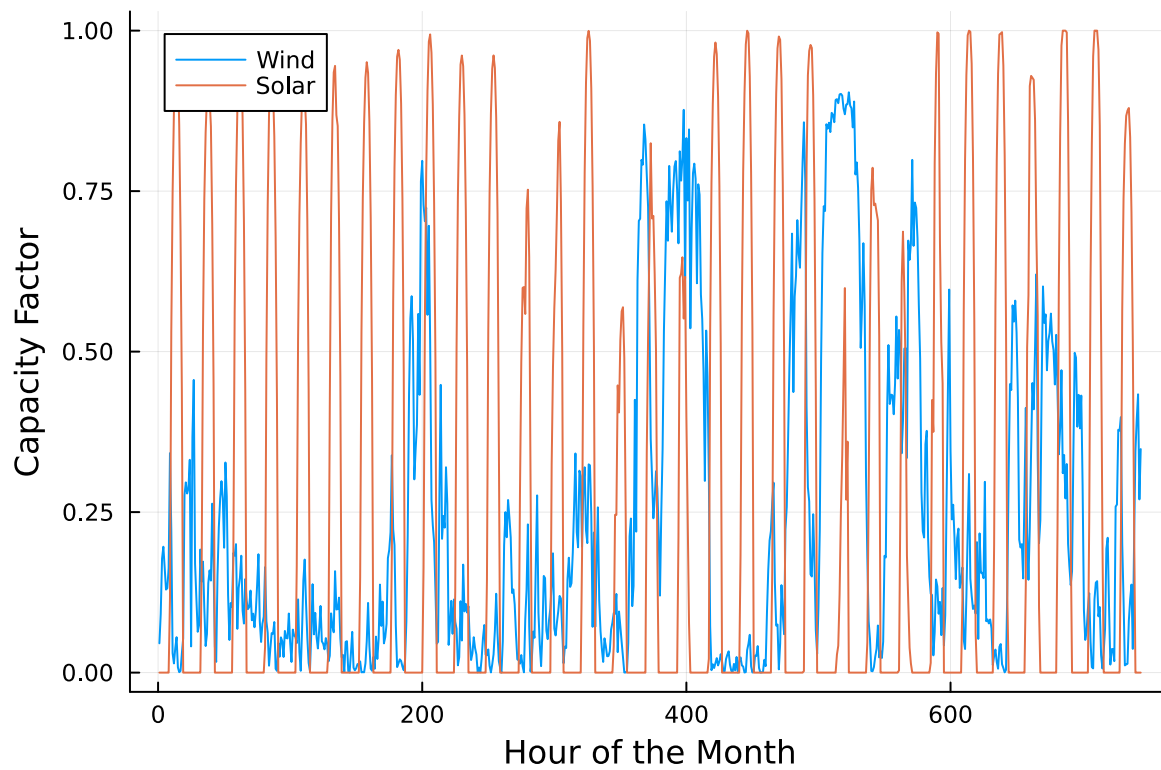
While coal, CCGT, and CT plants can generate at their full installed capacity, geothermal plants operate at maximum 85% capacity, and solar and wind available capacities vary by the hour depend on the expected meteorology. The utility will also penalize any non-served demand at a rate of \$10,000/MWh.

Problem 3.1

Formulate a linear program for this capacity expansion problem, including clear definitions of decision variable(s) (including units), objective function(s), and constraint(s) (make sure to explain functions and constraints with any needed derivations and explanations).

Problem 3.2

How much should the utility build of each type of generating plant? What will the total cost be? How much energy will be non-served?



Problem 3.3

What fraction of annual generation does each plant type produce? How does this compare to the breakdown of built capacity that you found in Problem 3.2? Do these results make sense given the demand and generator data?

Problem 3.4

Make a plot of the electricity price in each hour. Discuss any trends that you see.

Problem 3.5

What is the shadow price of CO₂? What is the value to the utility be of allowing it to emit an additional 1000 tCO₂/yr?

Significant Digits

Use `round(x; digits=n)` to report values to the appropriate precision! If your number is on a different order of magnitude and you want to round to a certain number of significant digits, you can use `round(x; sigdigits=n)`.

Getting Variable Output Values

`value.(x)` will report the values of a JuMP variable `x`, but it will return a special container which holds other information about `x` that is useful for JuMP. This means that you can't use this output directly for further calculations. To just extract the values, use `value.(x).data`.

Suppressing Model Command Output

The output of specifying model components (variable or constraints) can be quite large for this problem because of the number of time periods. If you end a cell with an `@variable` or `@constraint` command, I *highly* recommend suppressing output by adding a semi-colon after the last command, or you might find that your notebook crashes.

Problem 3 Solutions

Problem 3.1

Our decision variables:

C_i = Installed capacity of generator type i (MW)

$G_{i,t}$ = Electricity generated by plant type i at hour t (MWh)

NSE_t = Non-served demand at hour t (MWh)

Our objective functions: We want to minimize the cost of the system.

The cost of our system consists of the investment cost + operating costs = $\sum$ (Fixed Costs) (*installed capacity from each generator*) + $\sum \sum$ (Variable Costs) (*production from each generator*) + $\sum NSE_{cost} NSE_t$. It is given that the utility will penalize any non-served demand at a rate of \$10,000/MWh, which we can denote using P_{NSE} .

Thus,

$$\min Z = \sum (F_i)(C_i) + \sum \sum (V_i)(G_{i,t}) + \sum (P_{NSE})(NSE_t)$$

Our constraints:

Demand balance: Electricity demand at hour t is equal to non-served demand at hour t + electricity generated by plant type at hour t

$$\sum (G_{i,t}) + NSE_t = D_t$$

CO₂ constraint: The NY state legislature is planning to enact an annual CO₂ limit ($CO2_{max}$), which for the utility would limit the emissions in its footprint to 1.5 MtCO₂/yr.

$$\sum \sum E_i * G_{i,t} \leq CO2_{max}$$

Capacity constraint: While coal, CCGT, and CT plants can generate at their full installed capacity, geothermal plants operate at maximum 85% capacity, and solar and wind available capacities vary by the hour depend on the expected meteorology.

For coal, CCGT, and CT:

$$G_{i,t} \leq C_i$$

For geothermal:

$$G_{i,t} \leq 0.85C_i$$

For wind and solar: we need to account for capacity factor of generator i at hour t .

$$G_{i,t} \leq CF_{i,t} * C_i$$

Non-negativity constraint: decision variables cannot be negative.

$$C_i, G_{i,t}, NSE_t \geq 0,$$

```
# load files and reformat dates
NY_demand = DataFrame(CSV.File("data/2013_hourly_load_NY.csv"))
rename!(NY_demand, :Time Stamp => :Date)
demand = NY_demand[:, [:Date, :C]]
rename!(demand, :C => :Demand)
demand[:, :Hour] = 1:nrow(demand)
T = nrow(demand)
hours = 1:T

generators = DataFrame(CSV.File("data/generators.csv"))
plants = collect(gens.Plant)

capacity_factor = DataFrame(CSV.File("data/wind_solar_capacity_factors.csv"))

# retrieve parameters from csv
fixed_costs = Dict{String,Float64}()
variable_costs = Dict{String,Float64}()
emissions = Dict{String,Float64}()
for i in eachrow(generators)
    pname = String(i.Plant)
    fixed_costs[pname] = float(i.FixedCost)
    variable_costs[pname] = float(i.VarCost)
    emissions[pname] = float(i.Emissions)
end

# build dictionary of capacity factors for wind, solar, geothermal (85%), and coal/ccgt/ct (100%)
CF = Dict{String,Vector{Float64}}()
for i in plants
    if i == "Wind"
        CF[i] = cap_factor.Wind[1:T]
    elseif i == "Solar"
        CF[i] = cap_factor.Solar[1:T]
    elseif i == "Geothermal"
        CF[i] = fill(0.85, T)
    else
        CF[i] = ones(T)
    end
end
```

```

# Other parameters
NSE = 10_000.0 # $/MWh
CO2_max = 1.5e6 # tCO2/yr

# convert demand to Float64 (MWh)
Demand = [float(demand.Demand[t]) for t in hours]

# build model
model = Model(HiGHS.Optimizer)
set_silent(model)

@variable(model, Capacity_installed[i in plants] >= 0) # MW
@variable(model, Generation[i in plants, t in hours] >= 0) # MWh in hour t
@variable(model, Nonserved[t in hours] >= 0) # MWh

# demand constraint
demand_constraint = Vector{ConstraintRef}(undef, T)

@objective(model, Min,
    sum(fixed_costs[i]*Capacity_installed[i] for i in plants) +
    sum(variable_costs[i]*Generation[i,t] for i in plants, t in hours) +
    sum(NSE*Nonserved[t] for t in hours)
)

# demand balance constraints
for t in hours
    demand_constraint[t] = @constraint(model, sum(Generation[i,t] for i in plants) + Nonserved[t] == Demand[t])
end

# capacity constraints
capacity_constraint = Dict{Tuple{String,Int}, ConstraintRef}()
for i in plants
    for t in hours
        capacity_constraint[i,t] = @constraint(model, Generation[i,t] <= CF[i][t] * Capacity_installed[i])
    end
end

# CO2 constraint
co2_constraint = @constraint(model, sum(emissions[i]*Generation[i,t] for i in plants, t in hours) <= CO2_max)

optimize!(model)

total_cost = objective_value(model)

```

```

NSE_vals = [value(Nonreserved[t]) for t in hours]
total_nonserved_MWh = sum(NSE_vals)

# annual generation for each plant
annual_gen = Dict(i => sum(value.(Generation[i,t]) for t in hours) for i in plants)
total_generation = sum(values(annual_gen))

# fraction of generation versus capacity
fraction_generation = Dict(i => (annual_gen[i] / total_generation) for i in plants)
fraction_capacity = Dict(i => (Capacity_vals[i] / sum(values(Capacity_vals))) for i in plants)

# hourly prices: duals of demand constraints
hourly_prices = [dual(demand_constraint[t]) for t in hours] # $/MWh

# CO2 dual (shadow price $/tCO2)
co2_shadow = dual(co2_constraint)
value_1000t = co2_shadow * 1000.0

# display results
println("-----")
println("Problem 3.2")
println("-----")
for i in plants
    println("Installed capacity for ", i, " : ", round(Capacity_vals[i]; digits=2), " MW, cap")
end
println("The total cost is \$", round(total_cost, digits=2))
println("Total non-served energy (MWh/yr): ", round(total_nonserved_MWh; digits=2))

println("-----")
println("Problem 3.3")
println("-----")
for i in plants
    println(i, " generation: ", round(annual_gen[i]; digits=1),
           " MWh (", round(100*fraction_generation[i]; digits=2), "% of generation)")
end

println("-----")
println("Problem 3.5")
println("-----")
println("CO2 shadow price: \$", round(co2_shadow; digits=4), " per tCO2")
println("The value of allowing an additional 1000 tCO2/yr is \$", round(value_1000t; digits=2))

```

Problem 3.2

Installed capacity for Geothermal : 285.57 MW, capacity % = 3.99%
Installed capacity for Coal : 0.0 MW, capacity % = 0.0%
Installed capacity for NG CCGT : 1509.17 MW, capacity % = 21.11%
Installed capacity for NG CT : 703.06 MW, capacity % = 9.83%
Installed capacity for Wind : 2548.89 MW, capacity % = 35.65%
Installed capacity for Solar : 2103.75 MW, capacity % = 29.42%
The total cost is \$7.8535295894e8
Total non-served energy (MWh/yr): 332.29

Problem 3.3

Geothermal generation: 1.0969955e6 MWh (6.7% of generation)
Coal generation: 0.0 MWh (0.0% of generation)
NG CCGT generation: 3.3227666e6 MWh (20.3% of generation)
NG CT generation: 129473.4 MWh (0.79% of generation)
Wind generation: 6.4513419e6 MWh (39.42% of generation)
Solar generation: 5.3669963e6 MWh (32.79% of generation)

Problem 3.5

CO2 shadow price: \$-182.7719 per tCO2
The value of allowing an additional 1000 tCO2/yr is \$-182771.94

Problem 3.2

The utility should build the following installed capacities of each type of generating plant:
285.57 MW of geothermal, 0.0 MW of coal, 1509.17 MW of NG CCGT, 703.06 MW of NG CT,
2548.89 MW of wind, and 2103.75 MW of solar.

The total cost per year would be the combination of the total fixed costs per year =
\$7.8535295894e8.

The total non-served energy is 332.29 MWh/yr.

Problem 3.3

The fraction of annual generation that each plant type produces is below: Geothermal should
generate 1.0969955e6 MWh (6.7% of generation, capacity share = 3.99%).

Coal doesn't generate anything (0% of generation and capacity share).

NG CCGT generates 3.3227666e6 MWh (20.3% of generation, capacity share = 21.11%).

NG CT generates 129473.4 MWh (0.79% of generation, capacity share = 9.83%).

Wind generates 6.4513419e6 MWh (39.42% of generation, capacity share = 35.65%).

Solar generates 5.3669963e6 MWh (32.79% of generation, capacity share = 29.42%)

We can see here that the largest percentage of generation comes from solar and wind energy (33% and 39% respectively). For both solar and wind, the % of installed capacity is slightly lower than the % of generation (29% and 36% respectively), which makes sense because these variable renewables are cheaper to run (due to no fuel cost). NG CCGT generates 20%, with % of capacity 21%. NG CT generates a tiny percentage (1%), with % of capacity 10%. This is because NG CT is only used for a few hours when demand is high. Coal is expensive because of the fuel costs, so it is not used at all. Geothermal is also used a little.

Problem 3.5

The shadow price of CO₂ is -182.7719 dollars per tCO₂. Because the shadow price is negative, this means the total cost would actually increase if the CO₂ limit was relaxed. Allowing more emissions (like adding 1000 tCO₂) would actually increase the objective cost. The value of allowing an additional emission 1000 tCO₂/yr would actually cause the utility to lose -182771.94 dollars a year.

References

List any external references consulted, including classmates.

Thank you to TA Samantha for providing much-needed assistance on Problem 3—explaining the linear program and walking me through the decision variables, objective functions, other parameters, and the constraints. Professor Srikrishnan’s slides were also consulted, specifically “Shadow Prices and Capacity Expansion” as well as “Lab: Linear Programming”, to aid my understanding of JUMP analysis + optimization. For the coding aspect, I consulted [Jump.dev](#) documentation and [Julia packages](#) documentation to help me structure my code and understand the code function formats. Lastly, I had trouble retrieving parameters (plant name, fixed costs, variable costs, and emissions) from the generators CSV file., as I ran into debugging errors. I consulted ChatGPT, which provided me with the correct Julia syntax.